

- | | | |
|---|---|---------------------------|
| 6 | <p>(a) (i) particle must be moving
with component of velocity normal to magnetic field</p> | <p>M1
A1 [2]</p> |
| | <p>(ii) $F = Bqv \sin \theta$
q, v and θ explained</p> | <p>M1
A1 [2]</p> |
| | <p>(b) (i) face BCGF shaded</p> | <p>A1 [1]</p> |
| | <p>(ii) between face BCGF and face ADHE</p> | <p>A1 [1]</p> |
| | <p>(c) potential difference gives rise to an <u>electric</u> field
<i>either $F_E = qE$ (no need to explain symbols)</i>
<i>or electric field gives rise to force (on an electron)</i></p> | <p>M1

A1 [2]</p> |

Page 4	Mark Scheme	Syllabus	Paper
	GCE AS/A LEVEL – October/November 2012	9702	42

7 (a) induced e.m.f./current produces effects/acts in such a direction/tends M1